Research plans linking environment and history

Lesson 8, 90 minutes. Combining tree-ring evidence

Learning objectives

Separate evidence for dates, environment and causes.

Interpret agreement and disagreement among sources.

Develop a testable question and an evidence plan.

Three applications and a shared procedure

Application	Main focus	Shared checks
Dendroarchaeology	Objects, buildings, activity	Records, dating, comparisons, limits
Dendroecology	Trees, stands, disturbance	Records, dating, comparisons, limits
Dendroclimatology	Regional, seasonal climate	Records, dating, comparisons, limits

R30 Historic England

R52 Yonenobu & Eckstein (2006)

Observation, dating and interpretation

Observation: directly measured or inspected features.

Dating: placement by crossdating and calendar anchoring.

Interpretation: inference about cutting, disturbance, climate or activity.

Wood and archaeological time

Ring formation — **Felling** — **Processing / use** — **Deposition**

Reuse and repair can separate timber and building dates.

Check the evidence connecting these stages.

Conceptual stages, not measured dates for a particular object.

Linking climate and history

Coinciding changes can motivate a causal hypothesis.

Examine food, water, land use, institutions and other mediating processes.

Match region, season and temporal resolution.

R52 Yonenobu & Eckstein (2006)

Case: wooden containers and dated wooden slips

Maeda and colleagues (2024) compare Nara wooden containers and wooden slips.

Tree-ring dates are compared with written dates.

Read how different sources constrain feature chronology.

R55 Maeda et al. (2024)

Comparing outer-ring and written dates

What event does the written year describe?

What outer wood is preserved in the ring-dated object?

Examine association, duration of use and redeposition.

Reading Nabunken's public records

Connect images, measurement paths and numerical records.

Check sample identifiers, context, species and preservation.

Verify rather than assume correspondence with a paper's samples.

An ecological view of growth changes

Suppression may involve climate, insects or competition.

Combine scars, recruitment, mortality and management records.

Examine comparison trees and survivorship bias.

R35 Nowacki & Abrams (1997)

R45 Walker et al. (2014)

The scope of a climate reconstruction

State region, season, climate variable and units.

Check calibration, verification and response stability.

Test the links from local climate to broader social change.

R52 Yonenobu & Eckstein (2006)

Exercise A: interpret three timber dates

All examples are fictional. A building record mentions 1210.

Timber	Last preserved ring	Bark	Location
A	1187	Absent	Part 1
B	1199	Present	Part 2
C	1213	Absent	Part 3

8 minutes: claims, limits and additional evidence.

Synthetic teaching data / schematic, not empirical results.

Exercise A: the limits of the conclusion

A: timber use after formation of the 1187 ring; felling unresolved.

B: the 1199 ring formed; inspect season and interval before use.

C: wood formed through 1213; reassess its relation to the 1210 record.

Synthetic teaching data / schematic, not empirical results.

Sources of uncertainty

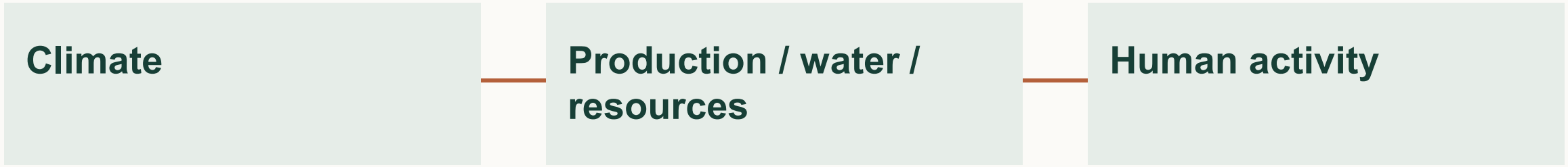
Stage	Questions to check
Measurement / dating	Boundaries, units, dating evidence
Preservation / sampling	Missing outer wood, biased samples
Interpretation	Post-felling intervals, alternative causes

R30 Historic England

R52 Yonenobu & Eckstein (2006)

Diagramming a causal hypothesis

Show proposed processes connecting climate, trees, production and activity.



Consider management, land use, institutions and other pathways.

Arrows indicate hypotheses, not proven causal effects.

Conceptual diagram. Connections are hypotheses to test.

R52 Yonenobu & Eckstein (2006)

Making a question specific

Specify place, period and material.

State competing hypotheses and predicted observations.

Identify evidence that could change the conclusion.

Planning sources and sampling

Check source location, preservation and access conditions.

Plan trees, sites, comparisons and sampling positions.

Prepare alternatives if the preferred material is unavailable.

Matching questions to methods

Calendar dating needs crossdating and an anchored reference.

Disturbance studies combine growth, injuries, populations and comparisons.

Climate estimation needs instrumental calibration and independent testing.

Tracing papers, reports and datasets

Move from the abstract to methods, materials, figures and limits.

Record dataset version, identifiers and the subset used.

Distinguish primary reports from introductory or secondary accounts.

R55 Maeda et al. (2024)

R56 奈良文化財研究所 / Nabunken (2024)

Presenting figures, data and sources

Make plots with clear axes, units and sample depth.

Distinguish measurements, synthetic data and schematics.

Check the applicable conditions and credits for external figures.

Exercise B: build your research plan

Write the question, sources, methods, comparisons, limits and additional evidence.

Item	What to include
Question and target	Place, period, competing explanations
Sources and methods	Materials, comparisons, dating, analysis
Limits and further evidence	Alternatives, observations that change the claim

8 minutes: draft 5, discuss 2, revise 1.

Reviewing research plans

Does the evidence address the question?

Are comparisons and independent checks included?

Are alternatives and unavailable-material contingencies considered?

Structuring the report

The syllabus indicates 2–4 A4 pages and at least three references.

Section	Purpose
Question	Target and reason for investigating
Sources and methods	Observation, dating and comparison
Discussion and conclusion	Evidence, alternatives and limits

Distinguish a proposal and expectations from completed results.

Conclusions and next checks

State the conclusion supported by the available evidence.

Identify remaining alternatives and uncertainty.

Name one further observation that could change the conclusion.

Keeping records for reanalysis

Preserve identifiers, original measurements, units and missing values.

Track the relationship among processing, figures, text and versions.

Clarify supplied materials and release scope in collaborative work.

Sources for continued learning

Use bilingual terms to search papers and data.

Connect Nabunken records, tree-ring databases and primary papers.

Continue testing explanations against observed material.

Final reflection

How do you distinguish a ring date from an event date?

Can you compare two explanations for one observation?

What new evidence could change your research conclusion?

Tree rings, environment and historical questions

Record observations, verify dates and compare explanations.

Move among specimens, data and literature.

Present a question together with the limits of its answer.